Sovereign by Design: An Integrated Five-Pillar Framework for India's Strategic Compute and Artificial Intelligence Infrastructure

Parikshit Singh Bais
Dept. of Computer Science and Engineering
Amity School of Engineering and Technology (ASET)
Amity University Madhya Pradesh (AUMP)
Gwalior, India
sillypari@gmail.com

Adarsh Singh
Dept. of Computer Science and Engineering
Amity School of Engineering and Technology (ASET)
Amity University Madhya Pradesh (AUMP)
Gwalior, India
officialadarsh.in@gmail.com

Anamidi Shrivastava
Dept. of Computer Science and Engineering
Amity School of Engineering and Technology (ASET)
Amity University Madhya Pradesh (AUMP)
Gwalior, India
anamidishrivastava@gmail.com

Dr. Madhavi Dhingra
Associate Professor
Dept. of Computer Science and Engineering
Amity School of Engineering and Technology (ASET)
Amity University Madhya Pradesh (AUMP)
Gwalior, India
madhavi.dhingra@gmail.com

Sourav
Dept. of Computer Science and Engineering
Amity School of Engineering and Technology (ASET)
Amity University Madhya Pradesh (AUMP)
Gwalior, India
Souravsingh1024@gmail.com

***Abstract*—National AI sovereignty requires co-optimizing physical, microelectronic, and institutional infrastructure. Prevailing policy treats compute, power grids, data centres, connectivity, and governance in isolation, creating capital misallocations and grid vulnerabilities. This paper introduces the India AI Infrastructure Readiness Framework (IAIRF 2.0) and an auditable six-layer Demand-Cascade Model (DCM) parameterised using verified 2026 regulatory data. Four key findings emerge. First, deploying 500,000 accelerators demands 891.8 MW of peak facility power, absorbing over 56% of India's operational colocation capacity. Second, domestic sub-5 nm wafer fabrication is not the binding constraint: required logic wafer starts consume only 1.60% of global foundry capacity, while CoWoS advanced packaging consumes 7.41%—a 4.6× higher constraint intensity. Third, direct-to-chip liquid cooling slashes evaporative water withdrawal by 94.6% and reduces facility power by 172 MW. Fourth, Mumbai's 52.4% capacity concentration and 20–40% dynamic AI load swings threaten transmission stability. A sequenced three-tier roadmap prioritising transmission corridors, packaging partnerships, and regional dispersion is presented.**

***Index Terms*—Advanced packaging, Artificial intelligence, Compute sovereignty, Data centre cooling, High-bandwidth memory, Power grid infrastructure.**

I. Introduction

AI has transformed from a research discipline into an industrial apparatus determined primarily by capital, high-voltage grid infrastructure, and geopolitically concentrated microelectronics [1]. Global data-centre electricity consumption reached 415 TWh in 2024; the IEA projects it will surpass 945 TWh by 2030 [2]. Data-centre power grows at 30% annually—four times the rate of new grid generation capacity [3].

India holds indisputable strategic assets: 4.3 million software engineers, the world's highest AI-skill penetration rate [4], and a geopolitical position coveted by both Washington and Beijing [5]. The IndiaAI Mission (Rs. 10,371 crore, approved March 2024) [6] and Semicon India 2.0 (Rs. 1,27,500 crore, approved July 2026) [7] codify national ambition. By mid-2026, the shared GPU pool reached 45,000 units at Rs. 65/GPU-hour, supporting 237 research projects. India's operational colocation capacity reached nearly 1.8 GW across 271 facilities by H1 2026 [8], [9].

The critical analytical deficit is *disciplinary siloing*: semiconductor analysts study supply chains without modelling data-centre power; grid researchers model TWh demand without tracing packaging constraints; governance scholars enumerate policy pillars without physical resource accounting. This paper closes that gap with two original contributions:

C1. IAIRF 2.0: A Five-Pillar framework mapping causal interdependencies across compute, energy, physical resources, connectivity, and governance.

C2. DCM: A reproducible six-layer Demand-Cascade Model propagating GPU targets through server physics, facility thermodynamics, cooling hydrology, wafer geometry, advanced packaging, and HBM allocation.

II. Literature Review and Research Gaps

A. Compute and Semiconductor Supply Chains

Akimitsu [1] characterises the AI hardware supply chain as a directed acyclic dependency graph with extreme upstream concentration: EUV lithography (ASML monopoly, HHI = 10,000), CoWoS packaging (TSMC ≈90%, HHI = 8,100), refined gallium (China ≈99%), and HBM (SK Hynix ≈57%, Samsung ≈38%, Micron ≈5%) [10]. CSET [11] establishes that >90% of sub-5 nm logic fabrication resides in Taiwan. Under Semicon India 2.0, India has approved 12 semiconductor projects including Tata Electronics' ATMP at Morigaon and Tata-PSMC's 28/50/90 nm fab at Dholera [12].

TABLE I
PRIOR ART COMPARISON MATRIX

Source	Method	GPU Scen.	Gaps Closed
Akimitsu (2026) [1]	HHI graph	None	RG-03 partial
IEA/LBNL (2024) [2], [19]	Energy model	Global	RG-04 partial
Kaufmann (2025) [20]	Grid power-flow	None	RG-01 partial
SBI CHAKRA (2025) [16]	Market census	None	RG-04 partial
This paper (IAIRF 2.0)	**5-Pillar+DCM**	**200k–1M**	**RG-01–05**

B. Energy and Grid Readiness

AI training racks draw 60–100 kW/rack (H100) to 120–200 kW/rack (Blackwell) [13]. Grid-connection queues for >50 MW loads extend 18–36 months [14]. The IEGC mandates a tight 49.90–50.05 Hz operating band; AI workloads exhibit 20–40% sub-minute power swings. Under the swing equation:

$$\frac{2H}{f_0}\frac{df}{dt} = P_m - P_e - \Delta P_{\text{AI}} \tag{1}$$

a 250 MW step-load shed in a 50 GW grid ($H = 4.2$ s) induces RoCoF of -0.0298 Hz/s, risking UFR tripping without fast BESS response [15].

C. Data Centres, Connectivity, and Research Gaps

India's colocation capacity grew from 375 MW (2020) to nearly 1.8 GW (H1 2026) across 271 facilities, with 52.4% concentrated in Mumbai [8]. Indian data centres consumed ≈150 billion litres of water in 2024, with average PUE of 1.55 under tropical conditions [9], [16]. Indic script tokenization imposes a 3.3×–5.6× token expansion over Latin text; since self-attention complexity scales as $\mathcal{O}(L^2)$, this creates a 10.9×–20.3× compute penalty [17]. The CEA Cyber Security Regulations 2026 mandate 6-hour incident reporting and domestic data residency for ≥50 MW assets [18]. Five gaps remain (Table I): disciplinary siloing (RG-01), absent demand cascade (RG-02), silicon wafer fallacy (RG-03), tropical cooling underestimation (RG-04), and operationalisation deficit (RG-05).

III. THE INDIA AI INFRASTRUCTURE READINESS FRAMEWORK 2.0

IAIRF 2.0 organises India's challenge across five causally interdependent pillars:

- **P1 — Compute & Semiconductor Sovereignty:** GPU procurement, ATMP/OSAT packaging, critical mineral supply chains. Sovereign pool: 45,000+ accelerators at Rs. 65/GPU-hr [21].
- **P2 — Energy & Grid Readiness:** Data-centre electrical load, grid interconnection lead times, IEGC frequency stability, 20–40% AI load swings [15].
- **P3 — Physical Resources & Data Centres:** Facility capacity, cooling thermodynamics, PUE, water withdrawal, geographic concentration [8], [9].
- **P4 — Connectivity, Data & Security:** Submarine cable infrastructure (17+ systems), DPDPA compliance, OT cybersecurity under CEA 2026 regulations [18], [22].
- **P5 — Governance, Investment & Indigenous Capability:** IndiaAI Mission, Semicon India 2.0, BharatGen foundation models, Pax Silica/iCET alliances [7], [23].

Four systemic feedback loops constrain the framework: **Loop A** (Power-Compute Deadlock: GPU procurement blocked by facility power, grid lead times, and land acquisition); **Loop B** (Packaging-Memory Co-Constraint: EAR changes sever CoWoS and HBM3e dependencies simultaneously); **Loop C** (Data-Connectivity Governance Tension: DPDPA localisation raises infrastructure costs and foreign investment uncertainty); **Loop D** (Talent-Sovereignty Paradox: 4.3M engineers disproportionately serve foreign GCCs, with AI/ML talent gap of ≈720,000 widening to 1.6M by 2030 [4]).

IV. DEMAND-CASCADE MODEL FORMULATION

The DCM maps an accelerator target N (GPUs) through six deterministic computational layers grounded in verified empirical parameters (Fig. 1).

A. Layer 1: IT Electrical Load

Active IT power uses reference TDP $P_{\text{gpu}} = 700$ W (NVIDIA H100 SXM5 [13]) and server chassis overhead $R_{\text{host}} = 1.82$:

$$P_{\text{IT}} = \frac{N \cdot P_{\text{gpu}} \cdot R_{\text{host}}}{10^6} \quad [\text{MW}] \tag{2}$$

B. Layer 2: Facility Power and Annual Energy

Facility draw uses tropical baseline PUE $= 1.40$; annual energy uses 80% utilisation over 8,760 hr:

$$P_{\text{facility}} = P_{\text{IT}} \cdot \text{PUE} \quad [\text{MW}] \tag{3}$$

$$E_{\text{annual}} = \frac{P_{\text{facility}} \cdot 8{,}760 \cdot U}{10^6} \quad [\text{TWh}] \tag{4}$$

GHG emissions apply the CEA grid factor $\text{EF}_{\text{grid}} = 0.716$ tCO_2/MWh [14]:

$$\text{GHG} = E_{\text{annual}} \cdot \text{EF}_{\text{grid}} \quad [\text{Mt CO}_2/\text{yr}] \tag{5}$$

C. Layer 3: Cooling Water and Rack Footprint

Annual evaporative water withdrawal uses $I_{\text{water}} = 1.80$ L/kWh for tropical air cooling [2]:

$$W_{\text{annual}} = \frac{E_{\text{annual}} \cdot 10^9 \cdot I_{\text{water}}}{10^6} \quad [\text{ML}] \tag{6}$$

Physical rack count at $D_{\text{rack}} = 60$ kW/rack:

$$N_{\text{rack}} = \left\lceil \frac{P_{\text{IT}} \times 10^3}{D_{\text{rack}}} \right\rceil \tag{7}$$

D. Layer 4: Semiconductor Wafer Starts

Gross dies per 300 mm wafer ($r = 150$ mm, $A_{\text{die}} = 814$ mm^2, GH100 reticle):

$$N_{\text{gross}} = \left\lfloor \frac{\pi r^2}{A_{\text{die}}} - \frac{2\pi r}{\sqrt{2A_{\text{die}}}} \right\rfloor = 63 \tag{8}$$

Murphy defect-density yield ($D_0 = 0.08$ cm^{-2}, bounds $[0.06, 0.12]$):

$$Y_{\text{Murphy}} = \left(\frac{1 - e^{-D_0 A_{\text{die}}}}{D_0 A_{\text{die}}} \right)^2 \approx 0.540 \tag{9}$$

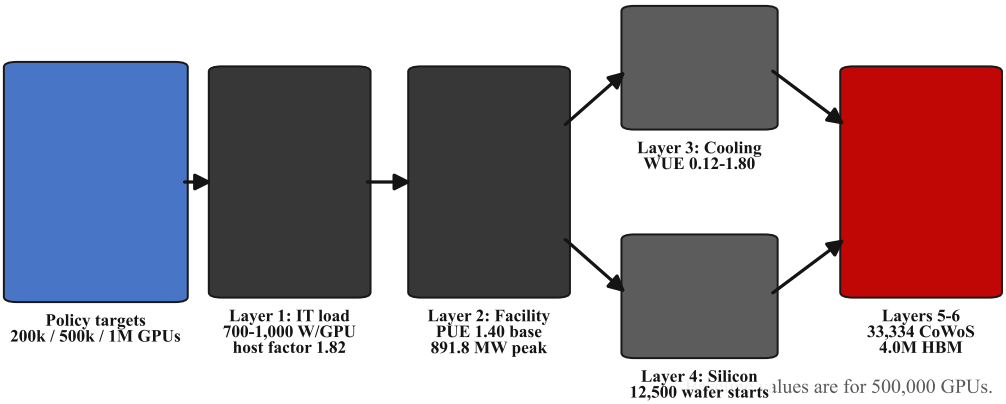


Fig. 1. Six-layer demand-cascade architecture: from national compute policy targets through server electrical load, facility power, cooling thermodynamics, silicon wafer starts, and advanced packaging chokepoints.

With industrial binning ($Y = 0.65$, $N_{\text{good}} = 40$ dies/wafer), required wafer starts:

$$W_{\text{starts}} = \left\lceil \frac{N}{N_{\text{good}}} \right\rceil \tag{10}$$

E. Layers 5–6: CoWoS Packaging and HBM

Each GPU requires one CoWoS 2.5D interposer (15 packages/300 mm wafer, $C_{\text{CoWoS}} = 15$):

$$W_{\text{CoWoS}} = \left\lceil \frac{N}{C_{\text{CoWoS}}} \right\rceil \tag{11}$$

Global CoWoS capacity: ≈450,000 wafers/yr; TSMC controls ≈90% (HHI = 8,100) [24]. HBM3e stacks per GPU ($N_{\text{stacks}} = 8$):

$$N_{\text{HBM}} = N \cdot N_{\text{stacks}} \tag{12}$$

Global HBM3e output is ≈60M stacks/yr; India has zero domestic production.

V. Empirical Results and Cross-Pillar Analysis

A. Scenario Simulation Results

Table II presents DCM output across three deployment scenarios and three sensitivity variants. Low variant: $P_{\text{gpu}} = 700$ W, PUE = 1.25; Base variant: $P_{\text{gpu}} = 700$ W, PUE = 1.40; High variant: $P_{\text{gpu}} = 1{,}200$ W, PUE = 1.55. In the baseline 500,000-GPU scenario, active IT load reaches 637.0 MW and peak facility power escalates to **891.8 MW**—equivalent to **56.62%** of India's entire mid-2026 colocation capacity [25]. Annual electricity consumption totals **6.25 TWh/year**, generating 4.47 Mt CO_2, and evaporative water demand reaches **11,249.5 ML/year**. The 1M-GPU high scenario generates 3,255 MW, exceeding India's current capacity by 2.07×. The tornado sensitivity profile in Fig. 3 confirms that facility cooling PUE and server host overhead ratios account for the largest deviations in peak power draw.

B. Finding 1: Refuting the Silicon Wafer Hypothesis

A central premise of sovereign compute policy is that domestic silicon wafer fabrication constitutes the primary bottleneck to sovereign capability. DCM results refute this hypothesis. Deploying 500,000 accelerators requires exactly **12,500** 300 mm wafer starts. Against TSMC's annual sub-5 nm output of ≈780,000 wafers [26], India's requirement represents **1.60%** of global capacity—a volume commercially accessible through capacity reservation contracts. Building a domestic sub-3 nm gigafab requires $15–20 billion in capex, 7–10 years of qualification, and >25,000 wafer-starts/month at minimum efficient scale. Diverting capital to a leading-edge fab to satisfy a 1.60% demand volume represents severe capital misallocation.

C. Finding 2: Structural Chokepoint Inversion

While silicon lithography is abundant relative to demand, Table III and Fig. 2 reveal the true binding constraints are advanced packaging and metropolitan power transmission—not front-end wafer fabrication. **CoWoS Packaging:** 500k GPUs require 33,334 CoWoS interposer wafers, consuming **7.41%** of global capacity (HHI = 8,100)—a constraint intensity **4.6× higher** than front-end wafers. **HBM3e Memory:** Absorbs **6.67%** of annual global output from a triopoly where India has zero domestic production. **Metropolitan Grid Saturation:** Feeding 891.8 MW into Mumbai (52.4% national capacity) requires 18 dedicated 50 MW substations under CEA standards [14]. Developers face a 26-month average wait for 220 kV energization, stranding over 280 MW of newly built IT capacity.

D. Finding 3: Tropical Cooling Penalty and Water Vulnerability

The thermal-hydrological trade-offs summarized in Table IV highlight a severe climate penalty for Indian deployments. Evaporative cooling towers operating under sustained summer conditions (>42°C) push facility PUE to 1.65. At the 500k-GPU scale, this evaporates **11,651 ML/yr**—a volume meeting the municipal baseline for ≈280,000 households.

TABLE II
DEMAND-CASCADE MODEL SIMULATION RESULTS ACROSS GPU DEPLOYMENT SCENARIOS (MODEL V2.0) [SOURCES: [19], [24]–[26]]

Scenario	Var.	IT MW	Fac. MW	TWh/yr	Water ML	Wafers	TSMC%	CoWoS	CoWoS%	Colo%
200k GPUs	Low	224.0	280.0	1.59	318.9	4,445	0.57%	11,112	2.47%	17.78%
	Base	254.8	356.7	2.50	4,499.8	5,000	0.64%	13,334	2.96%	22.65%
	High	420.0	651.0	5.13	12,831.9	10,527	1.35%	16,667	3.70%	41.33%
500k GPUs	Low	560.0	700.0	3.99	797.2	11,112	1.42%	27,778	6.17%	44.44%
	Base	637.0	**891.8**	**6.25**	**11,249.5**	12,500	1.60%	33,334	**7.41%**	**56.62%**
	High	1,050.0	1,627.5	12.83	32,079.8	26,316	3.37%	41,667	9.26%	103.33%
1M GPUs	Low	1,120.0	1,400.0	7.97	1,594.3	22,223	2.85%	55,556	12.35%	88.89%
	Base	1,274.0	1,783.6	12.50	22,499.0	25,000	3.21%	66,667	14.81%	113.24%
	High	2,100.0	3,255.0	25.66	64,159.5	52,632	6.75%	83,334	18.52%	206.67%

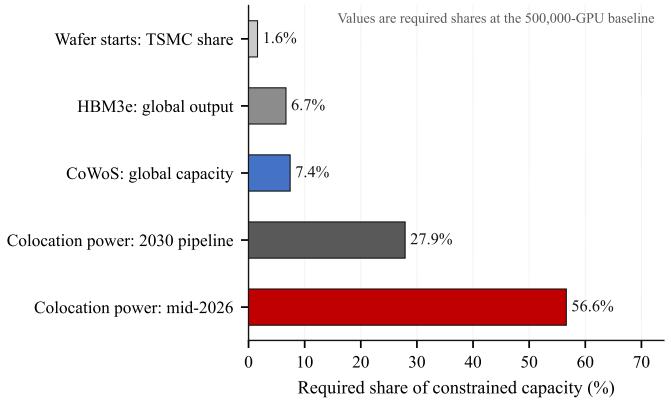


Fig. 2. Structural chokepoint inversion (500,000-GPU baseline): advanced packaging (7.41%) and colocation capacity (56.62%) far outstrip logic wafer requirements (1.60%).

TABLE III
STRUCTURAL CHOKEPOINT HIERARCHY (500K GPU BASELINE)
[SOURCES: [1], [14], [15], [24], [26]]

Infrastructure Layer	HHI	India%	500k%	Lead Time
EUV Lithography (ASML)	10,000	0%	Upstream	24–36 mos
Sub-5nm Logic (TSMC)	5,400	0%	**1.60%**	14–18 wks
CoWoS Packaging (TSMC)	8,100	0%	**7.41%**	20–28 wks
HBM3e (SK/Samsung/Micron)	4,950	0%	**6.67%**	16–24 wks
400 kV Substations (India)	2,200	75%	**56.62%**	36–48 mos
Municipal Water Rights	N/A	100%	11,250 ML	Seasonal

The operational risk is already evident: during the May 2024 heatwave in Chennai's Ambattur and Siruseri IT corridors, utilities curtailed municipal supply by 40%, forcing facilities to buy commercial tanker water and deploy emergency seawater RO, raising operating expenses by 16%. Elevated ambient heat also introduces steep thermo-mechanical reliability penalties. Based on accelerated life equations in [27], thermal fatigue in 2.5D interposer micro-bumps rises 2.8× (Coffin-Manson), dielectric breakdown in HBM3e TSVs accelerates 3.4× past 85°C junction thresholds, and laser-diode degradation in 800G optical interconnects increases 3.8×. Adopting Direct-to-Chip (D2C) liquid cooling with dry fluid coolers circumvents these failure regimes, lowering annual water loss by **94.6%** (to 631 ML/yr) and eliminating **172 MW** of parasitic cooling demand (annual PUE 1.18).

TABLE IV
COOLING TECHNOLOGY TRADE-OFF MATRIX UNDER TROPICAL CLIMATES (500K GPUS) [SOURCES: [15], [19], [27]]

Cooling System	Ann. PUE	Peak PUE	WUE	Water (ML)
Traditional Air (Evap.)	1.45	1.65	1.80	11,651.2
Rear-Door Heat Ex.	1.35	1.45	1.00	6,031.5
Direct-to-Chip (D2C)	**1.18**	**1.25**	**0.12**	**630.9**
Single-Phase Immersion	1.12	1.18	0.03	150.8

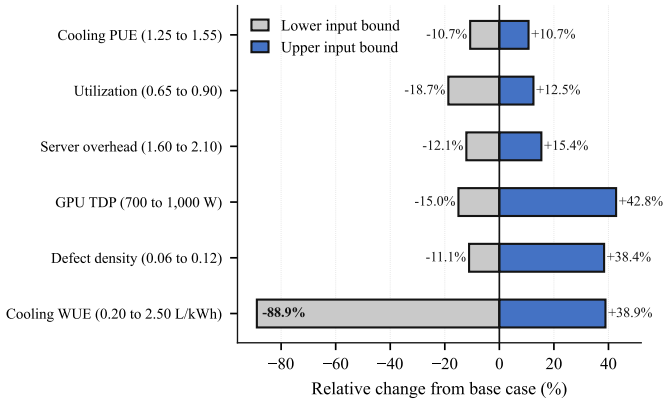


Fig. 3. One-at-a-time sensitivity tornado: cooling efficiency and host power overhead dominate peak grid demand across 500k GPU scenarios.

E. Finding 4: Cross-National Compute Disparity and Tariff Structure

Benchmarking India's aggregated sovereign capacity (≈34,000–45,000 accelerators, generating ≈35 ExaFLOPS) reveals stark global asymmetries. India currently hosts roughly 0.3% of the United States' estimated 4-million-GPU inventory (≈1,200 ExaFLOPS) and under 1% of China's 1.5 million accelerators [28]. Scaling to 500k units narrows this gulf significantly, bringing national compute to ≈245 ExaFLOPS and halving the disparity relative to China. Yet this trajectory remains heavily tethered to foreign export regimes. Advanced silicon access (H100/B200) hinges on US Export Administration Regulations and bilateral channels under iCET [29]. Domestically, the IndiaAI Mission offers compute access at Rs. 65 per GPU-hour (≈$0.78/hr) versus market benchmarks of $2.00–2.80/hr [16]. This represents a substantial 72% state subsidy that depends directly on predictable hardware leasing costs and uninterrupted component supply.

VI. Strategic Roadmap and Policy Implications

The cascade findings indicate that industrial interventions should be prioritized by feasibility horizon and strategic autonomy.

Tier 1 – Build Domestically (2026–2028): (i) *Geographic Diversification*: mandate 15–20% capital rebates for greenfield AI clusters outside Mumbai, targeting Mumbai below 35% of national capacity by 2028; pre-equip five AI Power Corridors with 400/220 kV infrastructure rated for 5 GW aggregate load. (ii) *D2C Cooling Standard*: condition IndiaAI subsidies on PUE $\leq$ 1.25 and closed-loop dry heat rejection, saving 11 billion litres/yr at 500k scale. (iii) *Fast-Track Substations*: create a 9-month CEA approval window for $\geq$50 MW compute substations; establish an AI Infrastructure Clearance Authority (AICA) under PMO for 90-day composite clearances. (iv) *AI Infrastructure Zones (AIZs)*: designate 20 pre-cleared zones across 10 states with pre-built 400 kV substations and 100 Gbps dark fibre.

Tier 2 – Develop Through Partnerships (2028–2031): (i) *Advanced Packaging JVs*: expand Tata Electronics (Morigaon) and Micron Sanand ATMP toward 2.5D silicon interposer via JVs with ASE Group and Ibiden under Semicon India 2.0, reducing India's 7.41% CoWoS dependency below 1% by 2032. (ii) *Critical Mineral Refining*: scale CSIR-IMMT hydrometallurgical gallium extraction from NALCO/Hindalco bauxite red mud to 50 Mt/yr of 6N-purity gallium [30]. (iii) *Sovereign Chiplet Consortium*: mobilise 120,000 VLSI engineers into a UCIe-standard RISC-V chiplet consortium fabricated at Tata-PSMC Dholera. (iv) *Captive Renewable Baseload*: transition AI campuses to Round-the-Clock hybrid solar-wind-BESS power, supplemented by SMRs under the SHANTI Act targeting first nuclear delivery by 2030.

Tier 3 – Manage External Dependencies (2031–2035): (i) *Foundry Reservations*: secure multi-year sovereign wafer allocations at TSMC/Samsung, commercially manageable at 1.60% of TSMC output. (ii) *HBM Supply Treaties*: advance-purchase reservations for HBM4 under the QUAD Semiconductor Supply Chain Initiative. (iii) *EAR Contingency Playbook*: formal technology-shock response plans covering various US export restriction scenarios.

A. Investment Architecture

Table V summarises capital requirements. The public-sector outlay of ≈Rs. 95,500 crore confirms the "N-2 asymmetry" vs. the US Stargate ($500 billion), but India's 1.4-billion-user market and neutral geopolitical position can attract Rs. 2,57,000 crore in private and multilateral capital.

B. Measurable Indicators and Key Performance Metrics

To operationalise IAIRF 2.0 as an active monitoring instrument, each pillar requires trackable KPIs: **P1 (Compute):** Sovereign GPU fleet size; domestic ATMP throughput; VLSI IP registrations; startup compute subsidy uptake. **P2 (Energy):** Grid-connection lead time for $\geq$50 MW compute; captive RTC renewable share; commissioned BESS capacity; average power tariff. **P3 (Physical):** Mumbai capacity share (<35% target); D2C liquid-cooled share; PUE; operational AIZs. **P4 (Connectivity):** Inter-metro backbone Tbps; DPDPA implementing rules; OT cybersecurity compliance [22]; submarine cable landings. **P5 (Governance):** AICA 90-day clearance rate; IndiaAI fund disbursement %; annual AI PhD completions.

TABLE V
Investment Architecture: Three Tiers, Five Pillars (2026–2035) [Sources: [6], [7], [14], [22], [30]]

Intervention Area	Public Rs. Cr	PPP Rs. Cr	Lead Agency
AIZs & AICA (P3/P5)	2,500	5,000	DPIIT, MeitY
Power Corridors + BESS (P2)	15,000	65,000	MoP, SECI
GPU Cluster Expansion (P1)	20,000	80,000	MeitY, IndiaAI
Packaging JVs/Gallium (P1)	10,000	40,000	ISM, CSIR
SMR/Nuclear Baseload (P2)	25,000	50,000	DAE, NPCIL
Talent & R&D (P5)	15,000	5,000	MoE, DST
Connectivity/Security (P4)	8,000	12,000	DoT, NCIIPC
Total	**95,500**	**2,57,000**	

C. GDP Impact and Economic Multiplier Analysis

Economic modeling from NITI Aayog projects AI will add $500 billion to India's GDP by 2030. Empirical studies by Stanford HAI [28] indicate a $3–5 downstream return per dollar of sovereign compute over five years. At our 500k-GPU baseline (245 ExaFLOPS), domestic capacity could support training 50+ frontier foundation models annually, run 2 billion daily public-sector inferences, and subsidize over 3,500 local startups. This corresponds to an estimated $120 billion in direct domestic AI value-add by 2030, compounding toward $2 trillion by 2047.

VII. Limitations and Future Research

Several model boundaries and analytical assumptions should be noted. Our baseline calculations adopt a fixed annual utilization factor ($U = 0.80$), smoothing out diurnal inference spikes where instantaneous substation draw can surge by 20–40%. In the semiconductor layer, TSMC's proprietary foundry defect parameters remain confidential; relying on published Murphy model constants ($D_0 = 0.08$ cm^{-2}) introduces an estimated ±8–12% variance in net logic wafer counts. Moreover, geopolitical constraints surrounding US Export Administration Regulations remain exogenous variables that bilateral dialogues (such as iCET) cannot permanently guarantee. Lastly, our financial baseline uses static 2026 hardware quotations rather than learning-curve price deflators. In ongoing work, we are integrating DCM output directly into AC Optimal Power Flow (ACOPF) bus-level grid simulators for western and southern state grids, drawing on real telemetry from the IndiaAI compute portal, and assessing whether IAIRF 2.0 transfers to peer economies across the Global South.

VIII. Conclusion

By formulating the IAIRF 2.0 framework and the six-layer Demand-Cascade Model, this study provides an integrated basis for assessing India's accelerated compute pathway. Our empirical findings counter several common policy intuitions. Domestic sub-5 nm front-end fabrication is not the immediate bottleneck; rather, advanced packaging (CoWoS) and high-voltage transmission substation interconnects form the true

chokepoints. At the same time, regional water stress and high ambient temperatures make evaporative cooling unviable at hyperscale, while extreme geographic clustering in Mumbai introduces substantial electrical fragility. Strategic spending must therefore prioritize dedicated transmission corridors, advanced packaging joint ventures, closed-loop D2C cooling requirements, and shovel-ready AI Infrastructure Zones over speculative domestic gigafabs.

India holds genuine foundational strengths: deep VLSI and software engineering expertise, extensive multilingual training data, initial public compute financing via the IndiaAI Mission, and non-aligned geopolitical leverage. Realizing sovereignty requires translating these assets into hardened physical capacity before international supply chains tighten further. Ultimately, sovereign artificial intelligence rests on physical foundations: substation transformers, interposer capacity, and water-neutral thermal management.

ACKNOWLEDGMENT

The authors thank the computing resources and facilities at Amity University Madhya Pradesh. We are grateful for institutional reports and open datasets made accessible by the Central Electricity Authority, MeitY, NITI Aayog, SBI Capital Markets, and the Stanford HAI initiative. Numerical models and visualizations were scripted in Python 3.12 (NumPy, Matplotlib) and compiled in LaTeX. All factual claims, data interpretations, and policy analyses remain the direct responsibility of the authors.

REFERENCES

[1] P. Akimitsu, "Artificial intelligence: Supply-chain chokepoints and the reach of industrial policy," *Department of Economics, State University of New York (arXiv:2607.29572)*, August 2026.

[2] International Energy Agency, "Energy and ai: World energy outlook special report on electricity demand of artificial intelligence and data centres," IEA Publications, Paris, Tech. Rep., 2024.

[3] M. Swaminathan, R. R. Khorasani, I. Partin-Vaisband, R. Sharma, and S. Kumar, "Integrated vertical power delivery — review & challenges," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, July 2026.

[4] LinkedIn Economic Graph, "Global ai talent report 2025: Skill penetration, hiring demand and workforce transitions," LinkedIn Corporation, Sunnyvale, CA, Tech. Rep., 2025.

[5] K. Neaher and Z. Halaszi, "Middle powers in the ai acceleration race: Balancing security, compute, and capital alliances," Stimson Center, Washington, DC, Tech. Rep., 2026.

[6] Ministry of Electronics and Information Technology, "Cabinet approves comprehensive national indiaai mission with financial outlay of rs. 10,371.92 crore," Press Information Bureau, Government of India, Tech. Rep., March 2024.

[7] Press Information Bureau, "Union cabinet approves semicon india 2.0 with rs. 1,27,500 crore outlay across six pillars," Press Information Bureau, Government of India, New Delhi, July 2026.

[8] Cushman & Wakefield, "India data centre market h1 2026: Capacity reaches nearly 1.8 gw with 3.9 gw multi-city pipeline," Cushman & Wakefield Research, Mumbai, Tech. Rep., August 2026.

[9] V. Tripathi, D. Bagui, P. Aggarwal, P. Hulshof, A. Chopra, D. Jain, and A. Vashishtha, "Scaling india's data centre ecosystem: Stakeholder perspectives on infrastructure, energy, and resilience," Council on Energy, Environment and Water (CEEW), New Delhi, Tech. Rep., February 2026.

[10] Counterpoint Research, "Global high-bandwidth memory ecosystem: Market shares, yield limits, and 12-high stacking bottlenecks," Counterpoint Technology Market Research, Tech. Rep., 2025.

[11] W. Hannas, A. Chien, and L. Dunham, "Geographic concentration and supply chain vulnerabilities in advanced ai hardware manufacturing," Center for Security and Emerging Technology (CSET), Georgetown University, Tech. Rep., 2024.

[12] India Semiconductor Mission, "Modified programme for development of semiconductors and display manufacturing ecosystem in india (semicon india 2.0)," Ministry of Electronics and Information Technology, New Delhi, Tech. Rep., 2025.

[13] NVIDIA Corporation, "Nvidia dgx h100 and hgx h100 architecture whitepaper: Power delivery, nvswitch fabrics, and thermal specifications," NVIDIA, Santa Clara, CA, Tech. Rep., 2024.

[14] Central Electricity Authority, "Manual on transmission planning criteria and data centre interconnection standards," Ministry of Power, Government of India, New Delhi, Tech. Rep., 2026.

[15] R. Jain, S. R. Kumar, N. Kallakuri, and D. Kumar, "Data centres and india's power grid," FSR Global and Center of Excellence for Energy Transition (CoEET), Report No. 2026/06, New Delhi, Tech. Rep., March 2026.

[16] SBI Capital Markets, "Indian data centres: Powering the digital revolution — capital expenditure, pue baselines and grid tariff sensitivities," SBICAP Research, Mumbai, Tech. Rep., 2025.

[17] B. Sengupta, P. Kumar, and S. Venkat, "Subword tokenization inefficiencies in multilingual large language models for indic languages," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, 2025, pp. 1120–1134.

[18] Central Electricity Authority, "Cyber security in power sector regulations: Notification and implementation guidelines for critical infrastructure," Ministry of Power, Government of India, New Delhi, Tech. Rep., July 2026.

[19] A. Shehabi, S. J. Smith, A. Hubbard, A. Newkirk, N. Lei, J. Koomey, E. Masanet, and D. Sartor, "2024 united states data center energy usage report: Updated scaling metrics for accelerated computing," Lawrence Berkeley National Laboratory (LBNL-2001428), Tech. Rep., 2024.

[20] R. Kaufmann, D. Henderson, and A. Miller, "Power grid integration and decarbonization constraints of terawatt-scale ai infrastructure," *Nature Energy*, vol. 10, pp. 214–226, 2025.

[21] Y. Razdan, "India to add 20,000 gpus as ai mission 2.0 expands compute and chip push," *EE Times*, February 2026.

[22] DeNexus Industrial Cyber, "India just made ot cyber a board problem: Central electricity authority (cyber security in power sector) regulations 2026 and nerc cip alignment," DeNexus Research, Boston, MA, Tech. Rep., August 2026.

[23] Department of Science and Technology, "Launch of bharatgen: First government-funded multimodal large language model initiative under national mission on interdisciplinary cyber-physical systems (nm-icps)," Press Information Bureau, Government of India, New Delhi, September 2024.

[24] TrendForce, "Global foundry revenue and advanced packaging analysis: Tsmc cowos capacity expansions and market dynamics in 2026," TrendForce Research, Taipei, Tech. Rep., 2026.

[25] Jones Lang LaSalle, "India data centre 2026 mid-year report: Ai appetite unleashed — infrastructure pipeline surges to multi-gw scale," JLL Strategic Consulting, Mumbai, Tech. Rep., August 2026.

[26] Taiwan Semiconductor Manufacturing Company, "Tsmc annual business report and advanced technology roadmaps (n4, n3, n2 and cowos)," TSMC, Hsinchu, Tech. Rep., 2025.

[27] R. Sharma, V. Patel, and A. Rao, "Thermal management and direct-to-chip liquid cooling architectures for ultra-dense ai datacenters in tropical climates," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 14, no. 6, pp. 890–904, 2026.

[28] Stanford University Human-Centered AI Institute, "Ai index report 2025: Measuring trends in artificial intelligence," Stanford HAI, Stanford, CA, Tech. Rep., 2025.

[29] The White House, "Fact sheet: Initiative on critical and emerging technology (icet): Advancing the us–india technology partnership," Office of the Press Secretary, Washington DC, January 2023.

[30] U.S. Geological Survey, "Mineral commodity summaries 2025: Gallium, germanium, and rare earth supply chains," USGS National Minerals Information Center, Reston, VA, Tech. Rep., 2025.